

MAHESH MADEY

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with **2+ years** at Siemens building **Java Spring Boot** backends, React frontends, and **GenAI-powered** systems. Developed REST APIs, event-driven microservices, and **LLM agentic workflows** from design through deployment. Collaborative across product, security, and platform teams; proficient in **Java, Python, TypeScript** and cloud-native architectures on **Azure** and **AWS**.

EXPERIENCE

Siemens Technology and Services Private Limited

Aug 2024 – Present

Bengaluru, India

Software Engineer — Full Stack & GenAI

Aug 2025 – Present

- Developed **REST APIs with Java Spring Boot** — layered service architecture, Spring Data JPA, and Hibernate ORM — adopted by **3+ engineering teams** and integrated into the company-wide platform release pipeline.
- Contributed **React + TypeScript** UI components with retry logic and error-state handling, reducing client-side error rates by **~30%** across three product surfaces.
- Implemented **GenAI agentic workflows** using Semantic Kernel and MCP (Model Context Protocol), enabling LLM-to-backend context exchange deployed via CI/CD on Azure.
- Added **LLM guardrail pipelines** — input validation, response filtering, output schema enforcement — eliminating prompt-injection vectors across all AI endpoints.
- Documented GenAI system runbooks across Agile sprints, cutting new-engineer ramp-up time by **~20%**.

Graduate Engineer Trainee — MLOps & Backend Integration

Aug 2024 – Jul 2025

- Developed **YOLOv8 CV pipelines** for factory safety monitoring via **NVIDIA DeepStream**, reaching **~90% detection accuracy** across 5+ live camera feeds.
- Containerised inference services with **Docker**; connected ML outputs to backend REST APIs, shrinking end-to-end integration time by **~40%**.
- Supported the **MLOps lifecycle** — data preparation, fine-tuning, validation, and benchmarking — across three model iterations in a distributed Agile squad.

Software Development Intern — AI Automation

Apr 2024 – Jun 2024

Aity, Inc | USA (Remote)

- Built **Python pipelines** for AI dataset generation and model fine-tuning, cutting manual processing time by **~35%**; implemented agentic workflows with **LangChain and CrewAI** and containerised all environments in Docker, reducing setup errors to zero.

EDUCATION

B.Tech in Computer Science (Big Data Analytics)

2020 – 2024

SRM Institute of Science and Technology, Chennai | GPA: 8.8/10.0 | 3rd Place, CINTELS Next Gen AI Ideathon (200+ participants)

CERTIFICATIONS

Salesforce Developer Virtual Internship

Jul 2023

SmartBridge x Salesforce Partner x AICTE | Apex, LWC & API, Flows & Security, Org Setup, VS Code + CLI, Developer Super Set

PROJECTS

Smart Financial Planner & Investment Recommendation Platform | Java 21 · Spring Boot 3 · PostgreSQL · Redis · Kafka · React · TypeScript · Gemini API

- Built a **Spring Boot 3 / Java 21** backend with **Spring Security**, Spring Data JPA, and Hibernate; ingests bank statements, categorises transactions, and computes financial health scores across multi-account portfolios.
- Built **Kafka-driven event pipelines** for async transaction processing; layered **Redis caching** on goal-tracking queries, cutting p95 latency under concurrent load.
- Wired **Gemini API** to back an AI financial assistant and investment recommendation engine; surfaced results via a **React + TypeScript** dashboard with real-time goal progress and analytics.

Multi-Document AI Assistant | Node.js · RAG · PostgreSQL · FAISS · Gemini LLM · SSE

- Developed **Node.js/Express** APIs for document ingestion; implemented a **RAG pipeline** with Gemini LLM, HuggingFace embeddings, and FAISS vector search — sub-second queries over a **1,000+ document corpus** with **SSE streaming** for 50+ concurrent users.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, SQL

Backend: Spring Boot 3, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices, Node.js, FastAPI

Frontend: React, TypeScript, SSE streaming, performance-optimised rendering

AI / ML & GenAI: LLM guardrails, RAG pipelines, LangChain, Semantic Kernel, MCP, YOLOv8, NVIDIA DeepStream, Gemini, OpenAI API, Hugging Face, CrewAI

Databases & Messaging: PostgreSQL, Redis, Kafka, MongoDB, FAISS, NoSQL design, query optimisation

Cloud & DevOps: Azure, AWS (EC2 · S3 · RDS · Lambda), Docker, Kubernetes (familiar), CI/CD, Git, Jenkins, ELK Stack, CloudWatch

Security & Auth: Spring Security, JWT, OAuth 2.0, OIDC, input validation, output schema enforcement